

**SEARCHING FOR A CURE** A technician at work at the ImmunoACT lab in Mumbai

## 5 FACTS ABOUT BIOTECH CURES FOR CANCER

- Every product goes through extensive safety and efficacy trials. Some mediums, such as lentiviral vectors, which are used in CAR-T therapy, have a decade-long good safety record
- They can save lives, giving hope to cancer patients where previously there was none
- Seventy per cent of the new drugs currently under development are biotechnology drugs. A majority of cancer medicines are now based on genetically-altered molecules
- Genetically modifying your immune cells does not change your entire DNA. Rather, research has found that since cancer is often a result of a genetic defect, the cure is also most likely to be in finding ways to detect, correct or overcome the specific defect
- Biotech companies undergo rigorous checks for quality control and safety in their laboratories before they are given the licence to manufacture or provide any therapy

studies. Most researchers are trying to find monoclonal antibodies that can be used to activate an NK cell immune response. In India, immuno-oncology start-up Zumutor has come up with ZM008, a monoclonal antibody that can be used to target prostate cancer cells. The company is in the process of filing an IND (investigational new drug application) with the FDA and getting into phase 1 human clinical studies this year. Zumutor has three such molecules in its development pipeline: ZM008, ZM012, and ZM014.

In cancers where immunotherapy drugs do not work because of an enzyme called DNA methyltransferase, which halts gene transcription in immune cells, researchers are experimenting with combination immunotherapy. In a recent study published in the *Journal for ImmunoTherapy of Cancer*, researchers in the UK found that immunotherapy drugs combined with a DNA hypomethylating agent halted cancer progression in more than a third of the patients in a Stage 1 trial. The DNA hypomethylating drug works by switching off the rogue enzyme, thereby increasing the efficacy of immunotherapy drugs.

### LIQUID DETECTION

Early diagnosis is one more area where biogenetics is helping improve the chances of cancer patients. The earlier a tumour is detected, the better the chances of recovering fully from it. So far, we have had invasive biopsies to identify specific features—DNA, RNA and protein expression—in cancer biomolecules. Often, it is impossible to obtain a sample, particularly if the tumour is located in a high-risk or inaccessible area, making accurate diagnosis a challenge. That is set to change with a range of new non-invasive genetic diagnostic tools. All one needs is a simple blood test.

Take breast cancer in India. According to the WHO, one woman is diagnosed with the malady every four minutes in the country, and one woman dies of it every 13 minutes. However, if detected in time, there is an almost 90 per cent chance of recovering fully from it. Mammography remains the most reliable tool for diagnosis, according to Dr Ramesh Sarin, a surgical oncologist with Apollo Hospitals in Delhi, but is not sensitive enough and can miss cancer in younger women. MRI of the breast is not available as widely and can